

PROJECT REPORT

MathCrypt

A Multi-Algorithm Password Encryption System

A Python Command-Line Application

Submitted by

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Project Type

Academic Python Project

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2. Introduction

MathCrypt is a Python-based academic project that demonstrates how multiple mathematical operations can be combined to transform a user-provided numerical input into an encoded output. The project is designed as a command-line application and focuses on applying fundamental Python programming concepts in a practical way.

The system uses mathematical properties such as factors, prime checking, prime factorization, digit sum, digit reversal, binary conversion, factorial of digits, and sign checking. The results of these operations are processed according to the program's encryption logic to generate the final encrypted output.

MathCrypt is primarily an educational demonstration of algorithmic thinking, conditional statements, loops, functions, lists, strings, arithmetic operations, and input/output handling in Python. It is not intended to replace modern cryptographic password-hashing systems.

3. Problem Statement

Basic mathematical operations are often taught independently, making it difficult for beginners to see how several programming concepts can work together in a single application. The problem addressed by MathCrypt is to design a simple command-line system that takes an input and applies multiple mathematical transformations before combining their results into an encoded output.

The project provides a practical way to demonstrate how individual algorithms can be organized into a multi-step processing system while keeping the implementation understandable to a beginner-level Python programmer.

4. Objectives

The main objectives of the MathCrypt project are:

- To develop a basic mathematical password-encryption-oriented application using Python.
- To implement and combine multiple mathematical algorithms in one project.
- To practice conditional statements, loops, functions, lists, strings, and arithmetic operations.
- To understand how input data can be transformed through sequential processing steps.
- To generate a final encoded output from the results of different mathematical operations.
- To strengthen logical thinking and problem-solving skills through practical implementation.

5. Technologies Used

- Python 3.x

- Command-Line Interface (CLI)

No external Python libraries are required for the basic implementation.

6. Methodology / Algorithm

MathCrypt follows a sequential processing approach. The input is first accepted from the user and is then passed through a series of mathematical operations. The individual results are collected and combined according to the encryption logic implemented in the program.

General Algorithm

1. Start the program.
2. Accept the required numerical input from the user.
3. Find the factors of the input.
4. Check whether the input is prime.
5. Find the prime factors of the input.
6. Calculate the sum of the digits.
7. Reverse the digits of the input.
8. Convert the input into binary.
9. Calculate the factorial of the individual digits.
10. Check the sign of the input.
11. Combine the generated results according to the encryption logic.
12. Display the final encrypted output.
13. Stop the program.

Mathematical Processing

Factors: Identifies numbers that divide the input without a remainder.

Prime Check: Determines whether the input has exactly two positive factors.

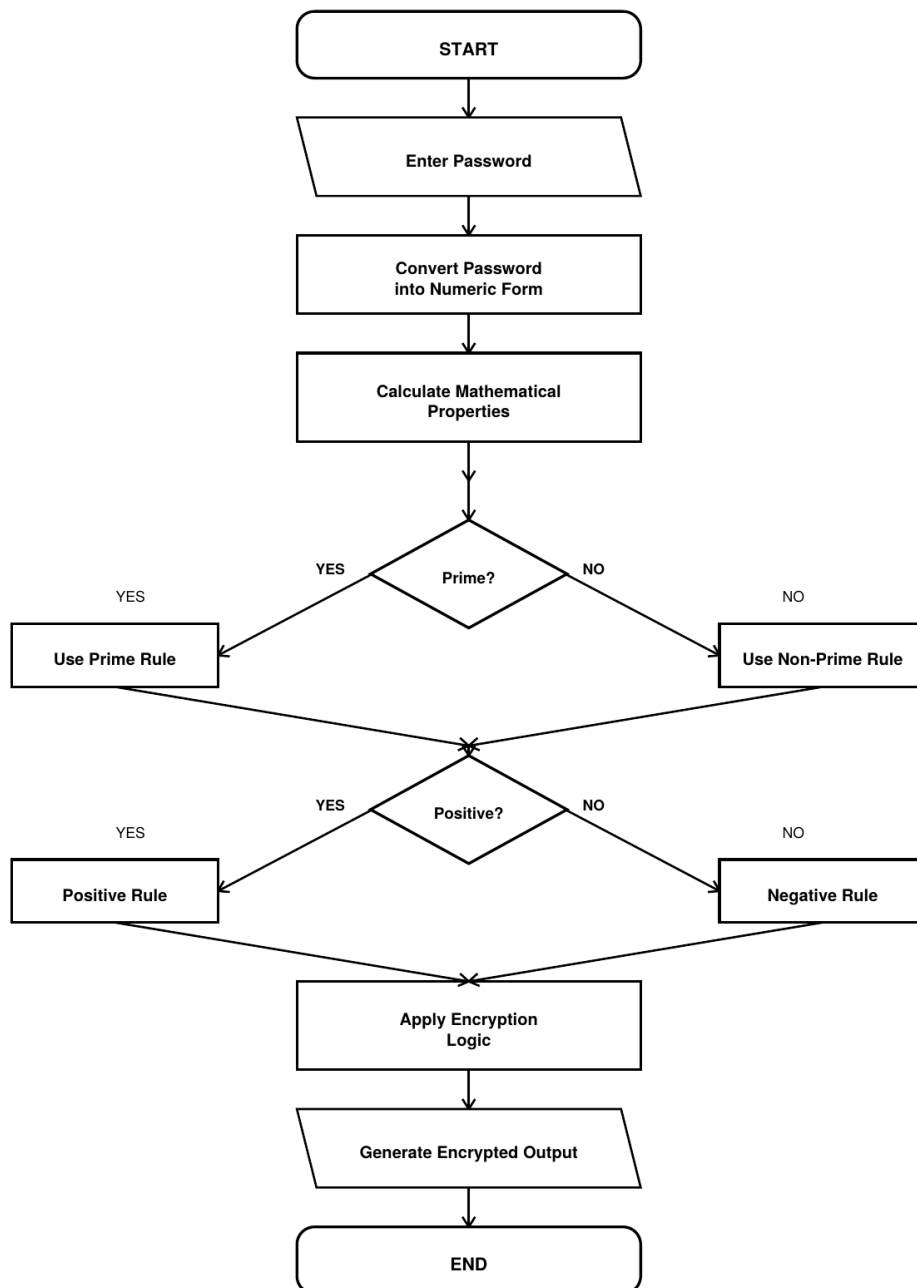
Prime Factors: Decomposes a composite input into its prime factors.

Digit Sum: Adds the individual digits.

7. Flowchart

MathCrypt – Compact Project Flowchart

A Multi-Algorithm Password Encryption System



Compact flow designed for clear single-page/slide presentation.

8. Implementation

The project is implemented as a Python command-line program. The implementation uses basic Python constructs so that each mathematical operation can be understood and tested independently.

Core implementation areas include:

- Input handling for receiving the user's value.
- Loops for processing digits and finding factors.
- Conditional statements for prime and sign checks.
- Lists for storing factors and prime factors.
- String operations for digit manipulation and formatted output.
- Arithmetic operations for sums, factorials, and other calculations.
- Binary conversion for representing the input in base 2.
- Final combination of intermediate results into an encrypted output.

GitHub Repository

<https://github.com/karthikcode0/MathCrypt-A-Multi-Algorithm-Password-Encryption-System>

9. Features

- User input through the command line.
- Factor calculation.
- Prime number checking.
- Prime factor calculation.
- Digit sum calculation.
- Digit reversal.
- Binary conversion.
- Factorial calculation for individual digits.
- Sign checking.
- Combination of multiple mathematical results.
- Generation of a final encrypted output.

10. Sample Input/Output

For an example input of 1234, the mathematical properties are:

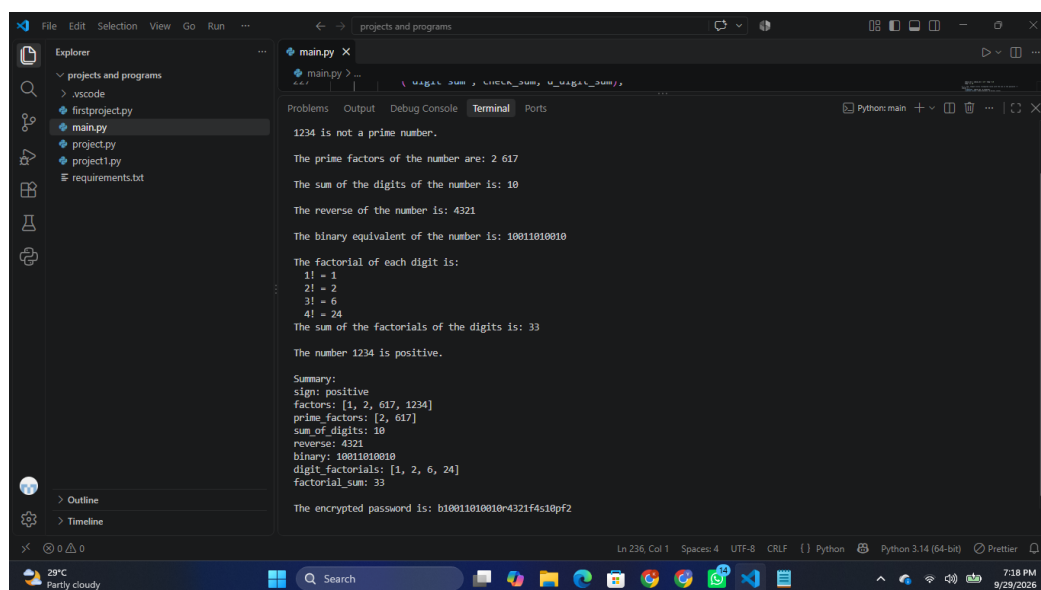
Property	Result
Input	1234
Factors	1, 2, 617, 1234
Prime Check	Not Prime
Prime Factors	2, 617
Digit Sum	10
Reverse	4321
Binary	10011010010
Factorial of Digits	1, 2, 6, 24
Sign	Positive

The exact final encrypted output is generated by the encryption logic implemented in the submitted source code.

11. Results

The MathCrypt project demonstrates the application of multiple mathematical operations to a single input. For the sample input 1234, the system derives its factors, determines that it is not prime, identifies its prime factors, calculates its digit sum, reverses its digits, converts it to binary, calculates the factorials of its individual digits, and determines its sign.

The intermediate results are then used by the program's encryption logic to create the final encoded output.



```
main.py > ...
1234 is not a prime number.

The prime factors of the number are: 2 617

The sum of the digits of the number is: 10

The reverse of the number is: 4321

The binary equivalent of the number is: 10011010010

The factorial of each digit is:
1! = 1
2! = 2
3! = 6
4! = 24

The sum of the factorials of the digits is: 33

The number 1234 is positive.

Summary:
sign: positive
factors: [1, 2, 617, 1234]
prime_factors: [2, 617]
sum_of_digits: 10
reverse: 4321
binary: 10011010010
digit_factorials: [1, 2, 6, 24]
factorial_sum: 33

The encrypted password is: b10011010010r4321f4s10pf2
```

Figure 1: Program output for the sample input 1234 (encrypted password: b10011010010r4321f4s10pf2)

12. Limitations

MathCrypt is an academic project intended to demonstrate mathematical transformations and Python programming concepts. Its mathematical combination method should not be treated as a secure replacement for established password-hashing or encryption systems used in real-world security applications.

The strength of the final encoded output depends on the mathematical combination implemented in the project.

13. Future Scope

- Add established cryptographic password-hashing techniques.
- Introduce secure random salt generation.
- Add password strength analysis.
- Support a graphical user interface.
- Add secure user authentication.
- Provide multiple encryption or hashing modes.
- Add secure file or database-based storage.
- Extend the project with additional mathematical transformations.
- Add automated testing for different input cases.

14. Conclusion

MathCrypt provides a practical demonstration of how fundamental mathematical operations and Python programming concepts can be integrated into a single project. By processing an input through factors, prime checking, prime factorization, digit sum, reversal, binary conversion, factorial calculations, and sign checking, the project illustrates a multi-stage transformation process.

The project provides useful practice in algorithm design, problem solving, functions, loops, conditional logic, lists, strings, and command-line input/output. Although it is not intended for production-level password security, it serves as an academic demonstration of combining multiple algorithms to produce an encoded result.

15. References

- Python Software Foundation. Python 3 Documentation.
- Python Software Foundation. Built-in Types and Built-in Functions.
- Python Software Foundation. Math — Mathematical Functions.
- GitHub repository: MathCrypt — A Multi-Algorithm Password Encryption System.
- The submitted source code and project implementation are the primary references for the specific algorithms and encryption logic used in this project.